

Chapter 78 — Surgical, Radiographic, and Endoscopic Anatomy of the Retroperitoneum

DEFINITION, BOUNDARIES AND ORGANS

- **Why it matters:** safe entry into the retroperitoneum underpins open, laparoscopic and robotic surgery and percutaneous access for endoscopy.
- **Etymology:** *retro* ("behind") + *peritoneum* (the lining of the abdominal cavity), so it is the space **behind the peritoneum**.
- **Boundaries:** **anterior** is the posterior reflection of the peritoneum; **posterior** is the abdominal wall; **cranial** is the diaphragm; **caudal** is the extraperitoneal pelvic structures.
- **Boundaries are debated:** there is no consensus on whether psoas and quadratus lumborum are boundaries or contents.
- **Retroperitoneum ≠ extraperitoneal space.** The extraperitoneal space is the retroperitoneum **plus the space circumferentially surrounding the abdominal cavity**.

PRIMARILY VERSUS SECONDARILY RETROPERITONEAL ORGANS

	Primarily retroperitoneal	Secondarily retroperitoneal
Development	Form within the retroperitoneum and stay there throughout	Begin within the mesentery . The peritoneum regresses and fuses with the posterior abdominal wall
Organs	Adrenals, kidneys, ureters , abdominal aorta, inferior vena cava (IVC), anal canal	Ascending and descending colon , duodenum (except the first part), pancreas (except the tail)
Space occupied	Perirenal space (the genitourinary organs)	Anterior pararenal space

BOX 78.1 ORGANS AND STRUCTURES OF THE RETROPERITONEUM

Category	Contents
Organs	Kidneys, ureters, adrenals (primary). Portions of the duodenum, ascending colon, descending colon and pancreas (secondary)
Vessels and lymphatics	Abdominal aorta and branches, IVC and tributaries, ascending lumbar veins, portal vein, lumbar lymph nodes, lumbar lymphatic trunks, cisterna chyli
Nerves	Branches of the lumbosacral plexus, sympathetic trunk, autonomic plexuses, autonomic ganglia

Data from Mirilas and Skandalakis, 2009.

- **What the box is for:** it sorts contents by system. The **primary/secondary tag on each organ** is the examinable part.
- **Trap:** the **first part of the duodenum** and the **tail of the pancreas** are the exceptions and are **not** retroperitoneal.

TABLE 78.1 ANATOMIC AND SURGICAL HISTORY OF THE RETROPERITONEUM

Theme	Contributors (year): contribution
Retroperitoneal tumours	Morgagni (1761): lipoma at autopsy. Lobstein (1829): first used the term "retroperitoneal tumour" . Broca and Moynier (1850): tumours at autopsy. Dickinson (1871): teratomatous tumour resembling ovarian dermoid. Bassini (1889): cystadenoma resembling ovarian pseudomucinous cystadenoma
Fasciae	Cloquet (1817): perirenal fascia. Treitz (1853): theory of "absorption", retroduodenopancreatic fascia. Toldt (1879, 1893): conjoined visceral fasciae. Zuckerkandl (1883): posterior renal fascia. Gerota (1895): anterior renal fascia

Theme	Contributors (year): contribution
Spaces and compartments	Bogros (1823): surgical anatomy of the iliac area. Rogie (1894): retroperitoneal anatomy. Poirier et al. (1923): lobulation of pararenal and perirenal fat. Drouet (1941): subperitoneal area. Baumann (1945): embryology of the renal area. Altemeier and Alexander (1961): extraperitoneal compartments above the pelvic brim. Stevenson and Ozeran (1969): extraperitoneal pelvis divided into posterior, anterior, inferior and superior spaces. Meyers et al. (1972): anterior and posterior pararenal and perirenal spaces
Imaging	Hureau et al. (1990, 1991): computed tomography (CT) of extraperitoneal spaces. Korobkin et al. (1992): CT of anatomy and fluid collections
Endoscopic surgery	Wickham (1979): pneumoretroperitoneum for endoscopic removal of a ureteric stone. Gaur (1992) and McDougall et al. (1994): retroperitoneal videoscopic renal surgery

Data from Skandalakis and Colborn, *Skandalakis' Surgical Anatomy*, 2004.

- **Pattern:** the sequence runs **autopsy tumours → fascial anatomy → named spaces → CT → minimally invasive surgery**.
- **Trap:** the eponyms are split. **Gerota described the anterior** renal fascia and **Zuckerkandl the posterior**, but Gerota's name is given to both laminae together.

Figures

- **Fig. 78.1, retroperitoneal spaces:** the diagram shows the **anterior pararenal space (APRS)**, **perirenal space (PR)** and **posterior pararenal space (PPRS)**, with renal fascia (RF), transversalis fascia (TF) and thoracolumbar fascia (TLF). The renal fascia is shown as **three laminae (anterior, middle, posterior)**. The **space of Bogros** is the caudal extension towards the pelvic floor and bladder. Modified from Skandalakis and Colborn, 2004.
- **Fig. 78.2 A–D, dissected retroperitoneum:** Bassett anatomic collection.
 - **Panels A and B** have the anterior Gerota's fascia removed. They show the coarser **pararenal fat** as distinct from the **perinephric fat**, the celiac and aortic autonomic plexuses, the inferior mesenteric ganglion, the lumbar nodes, and **psoas in its sheath**.
 - **Panels C and D** have the kidneys and adrenals sectioned and the IVC excised. They show the celiac ganglion, the **right renal vein cut short**, and **transversus abdominis with transversalis fascia** behind the kidney.

Concept — Every organ's position follows from **whether it formed behind the peritoneum or fused there later**. Secondly retroperitoneal organs sit in front of the kidney, in the anterior pararenal space, and that is why colon, duodenum and pancreas lie in the surgeon's path to the kidney.

BODY SURFACE LANDMARKS

- **Surface anatomy** of the kidneys is estimated from the **bony posterior abdominal wall** and guides diagnosis and incision planning.

RENAL SURFACE LANDMARKS

Landmark	Left kidney	Right kidney
Upper pole	11th rib	12th rib (right lies lower)
Hilum	L1	
Lower pole	L3–L4	

- **Fig. 78.3:** posterior projection of kidneys and ureters onto ribs 11–12, the lumbar transverse processes and L1 (female specimen). From *Gray's Anatomy for Students*, 2010.

POSTERIOR ABDOMINAL WALL

Flank Muscles

Superficial → deep:

External oblique (beneath subcutaneous fascia) arises from **ribs 5–12**. Fibres run **inferomedially** to the iliac crest and linea alba. The **inferior border of its aponeurosis forms the inguinal ligament**.

→ **Internal oblique** arises from the lumbodorsal fascia and iliac crest. Fibres run **superomedially** to the lower ribs and linea alba.

→ **Transversus abdominis**, named for its **transverse fibres**.

→ **Transversalis fascia** crosses the midline anteriorly and **fuses with the lumbodorsal fascia posteriorly**.

- **Each muscle layer is invested in its own fascia.** Together the layers **flex, extend and rotate the trunk and compress the abdominal contents**.
- **Examinable contrast:** external oblique fibres run **inferomedially** (“hands in pockets”), internal oblique fibres run **superomedially**.

Psoas, Iliacus, Quadratus Lumborum and Erector Spinae

- **Posterior compartment:** psoas, iliacus and quadratus lumborum are sometimes grouped as the posterior compartment of the retroperitoneum.
- **Psoas major:** arises from **T12–L5**, passes along the pelvic brim **behind the inguinal ligament** to the **lesser trochanter**. It **flexes the hip**. Nerve supply is the **anterior rami of L1–L3**.
- **Psoas minor:** **may be absent**. Arises from T12–L1 and inserts at the pelvic brim and iliopubic eminence.
- **Iliacus:** arises from the iliac fossa and lateral sacrum and inserts at the **lesser trochanter**. It **flexes the hip together with psoas major**.
- **Quadratus lumborum:** lies **posterolateral to psoas**. It runs from L5 and the **iliac crest** to the **12th rib** and the L1–L4 transverse processes. It **stabilises the 12th rib** and bends the trunk laterally.
- **Erector spinae (sacrospinalis):** a large group of back muscles that **extend the spine**.

TABLE 78.2 MUSCULATURE OF THE POSTERIOR AND LATERAL ABDOMINAL WALL

Muscle	Origin	Insertion	Function
Anterolateral flank group			
External oblique	Ribs 5–12	Lateral lip of iliac crest; aponeurosis to linea alba	Compresses abdomen; flexes trunk
Internal oblique	Lumbodorsal fascia, iliac crest, inguinal ligament	Lower four ribs; linea alba; pubic crest	Compresses abdomen; flexes trunk
Transversus abdominis	Lumbodorsal fascia, medial lip of iliac crest, ribs 7–12	Linea alba; pubic crest	Compresses abdomen only
Posterior compartment and back			
Psoas major	T12–L5	Lesser trochanter	Flexes hip
Psoas minor	T12–L1	Pelvic brim, iliopubic eminence	Weak flexion of lumbar spine
Iliacus	Iliac fossa, sacrum	Lesser trochanter	Flexes hip
Quadratus lumborum	L5, iliac crest	L1–L4 transverse processes, 12th rib	Depresses and stabilises 12th rib; lateral bending

Muscle	Origin	Insertion	Function
Erector spinae	Sacrum, vertebrae	Lower ribs, vertebrae	Extends spine

Modified from Drake, Vogl and Mitchell, *Gray's Anatomy for Students*, 2005.

- **What the table is for:** it separates the **flank layers**, which compress the abdomen, from the **posterior compartment**, whose muscles move the hip and spine.
- **Pattern:** **psoas major and iliacus share the lesser trochanter** and act together as hip flexors. **Transversus is the only flank layer that does not flex the trunk.**
- **Trap:** **psoas minor is inconstant** and inserts at the pelvic brim, not the femur.

Spine

- **Vertebral count:** **7 cervical, 12 thoracic, 5 lumbar**, then sacrum and coccyx.
- **Vertebral structure:** each vertebra has a weight-bearing body and a posterior and lateral arch forming the vertebral foramen. The spinous process projects **posteroinferiorly** and the transverse processes **posterolaterally**.
- **Lumbar vertebrae matter most here:** they are the largest and have **long, thin transverse processes**.
- **Vertebral level ≠ spinal cord segmental level:** the **sacral cord segments begin at vertebral T12–L1** in adults. In spinal cord injury, always **specify which level is meant**.

10th, 11th and 12th Ribs

- **Protective role:** the lower ribs shield the retroperitoneum. A **lower rib fracture → high suspicion of retroperitoneal injury**.
- **Shape:** lower ribs are **shorter and less angulated** than the upper ribs.

THE LOWER RIBS COMPARED

Rib	Neck / transverse process articulation	Angle	Sternal connection
10th	Head → vertebral body; neck → transverse process	Less pronounced than upper ribs	Not floating
11th	No neck ; no transverse process articulation	Less pronounced	None, floating rib
12th	Inferior border tethered to L1–L2 transverse processes by the costovertebral (lumbocostal) ligament	None ; shortest rib	None, floating rib

- **Dividing the costovertebral (lumbocostal) ligament** frees the 12th rib and gives **better upper retroperitoneal exposure in posterior approaches**. Dividing the **intercostal ligament** between other ribs does the same.
- **Floating ribs** (11th, 12th) are the palpable landmarks for **marking incisions**.
- **Intercostal neurovascular bundle** (Fig. 78.11): it runs between the **internal and innermost intercostal muscles** in the **costal groove on the lower margin of the rib above**. The order from top down is **vein → artery → nerve**. The **nerve is lowest and often unprotected** by the groove.
- **Percutaneous access and the ribs:** **supra-12th rib access may cross the pleural reflection**, causing symptomatic pleural puncture. **Supracostal access has more complications than subcostal**. Fig. 78.12 shows a supracostal left upper pole on fluoroscopy and retrograde pyelography in the prone position.

Figures

- **Figs. 78.4–78.6, posterior wall dissections:**
 - **Superficial (78.4):** latissimus dorsi, serratus posterior inferior, external and internal oblique, and the aponeurosis of transversus, with the right kidney outlined.
 - **Intermediate (78.5):** the **three layers of the lumbodorsal fascia**, sacrospinalis and quadratus lumborum in section.
 - **Deep (78.6):** the **costovertebral ligament** arising from the lumbar transverse processes, and the **kidney–pleura relationship**.
- **Fig. 78.7, flank layers in transverse section:** skin → superficial fascia, with a **fatty layer (Camper)** and a **membranous layer (Scarpa)** → external oblique → internal oblique → transversus → **transversalis fascia** → extraperitoneal fascia → parietal peritoneum. From *Gray's Anatomy for Students*, 2010.
- **Fig. 78.8, muscles of the posterior abdominal wall:** psoas major and minor, iliacus, quadratus lumborum and transversus, with the lumbar vessels. From *Gray's Anatomy for Students*, 2010.

Concept — *The posterior wall is a **layered barrier that each approach crosses differently**. Flank incisions cut muscle, while **posterior approaches exploit the 12th rib and its ligament**. Percutaneous access is limited above by the **pleura and the intercostal nerve**, and that is why subcostal access is safer.*

LUMBODORSAL FASCIA

- **Structure:** the lumbodorsal (thoracolumbar) fascia has **three layers** investing the posterior wall muscles, and they **merge into one laterally**.
- **Common access point:** the **tip of the 12th rib**, where all three layers have fused. There the single layer **merges anterolaterally with the transversus abdominis aponeurosis**.

THE THREE LAYERS OF THE LUMBODORSAL FASCIA

Layer	Medial attachment	Relation
Posterior	Spinous processes	Covers erector spinae
Middle	—	Separates erector spinae from quadratus lumborum
Anterior	Transverse processes	Covers the ventral surface of quadratus lumborum; continuous with the psoas fascia

- **Dorsal lumbotomy** enters the retroperitoneum **without cutting muscle**. It is a **vertical incision through the lumbodorsal fascia lateral to erector spinae and quadratus lumborum** (Fig. 78.14, from Anson and McVay, 1994). The American Urological Association (AUA) Core Curriculum on Surgical Incisions covers incision choice.
- **Percutaneous renal access** crosses the flank muscles, retroperitoneal fat and perirenal fascia. It is used for nephrostomy, renal biopsy and endoscopy under **fluoroscopic or ultrasound guidance**. Fig. 78.16 shows ultrasound of the tract to a **medullary pyramid**; access technique is covered in Chapter 98.
- **Fig. 78.13:** the lumbodorsal fascia layers in relation to the deep back muscles, psoas, quadratus lumborum and latissimus dorsi. From *Gray's Anatomy for Students*, 2010.

KEY POINTS: ORGANS AND BOUNDARIES OF THE RETROPERITONEUM

- **Contents:** kidneys, ureters, adrenals, pancreas, parts of the duodenum, ascending and descending colon, mesentery, vessels, lymphatics and nerves.
- **Boundaries:** **anterior** posterior peritoneal reflection, **posterior** abdominal wall, **cranial** diaphragm, **caudal** extraperitoneal pelvic structures.

- **Intercostal bundle** lies between the internal and innermost intercostals in the **costal groove of the rib** above.
- **Lumbodorsal fascia** is **three-layered** and **merges anterolaterally** with transversus abdominis.

Concept — Three fascial layers fuse into one at the **12th rib tip**. That single fused point is both the **standard flank access site** and the reason a **dorsal lumbotomy** can reach the kidney between muscles rather than through them.

RETROPERITONEAL FASCIAE AND SPACES

- **Embryology:** mesoderm-derived mesenchyme forms subcutaneous, body and retroperitoneal layers. The **retroperitoneal layer forms three strata** in late fetal life, and these strata **define the fasciae that compartmentalise the retroperitoneum**.

THREE STRATA → THREE FASCIAE → THREE SPACES

Stratum	Fascia formed	Associated organs	Space it bounds
Outer	Covers muscle epimysium → transversalis fascia	Abdominal wall	Posterior pararenal (posterior and lateral walls)
Intermediate	Renal (Gerota's) fascia: anterior and posterior laminae	Genitourinary	Perirenal
Inner	Fusion fascia with the primary dorsal peritoneum	Gastrointestinal	Anterior pararenal

Transversalis Fascia and Posterior Pararenal Space

- **Transversalis fascia** (outer stratum) lies **deep to transversus abdominis** and superficial to the preperitoneal fat and peritoneum. Behind the kidney it stays **anterior to the fascia of quadratus lumborum and psoas** (Fig. 78.18).
- **Medial fusion with the posterior lamina of Gerota's fascia** is clinically important. This fusion **must be incised to reach the renal hilum** in retroperitoneal dissection.

BOUNDARIES OF THE POSTERIOR PARARENAL SPACE

Boundary	Formed by
Anterior	Posterior lamina of Gerota's fascia
Posterior and lateral	Transversalis fascia
Medial	Fusion of transversalis fascia with the posterior lamina of Gerota's fascia

Gerota's Fascia (Renal Fascia) and Perirenal Space

- **Origin and name:** the laminae come from the **intermediate stratum**, which embeds the genitourinary organs. Together they form the **renal fascia**, named after **Dimitrie Gerota** (1867–1939), a Romanian anatomist.

ANTERIOR VERSUS POSTERIOR LAMINA OF THE RENAL FASCIA

	Anterior lamina	Posterior lamina
Eponym / synonym	Fascia of Toldt ; prerenal fascia	Fascia of Zuckerkanl ; retrorenal fascia
Thickness / imaging	Thinner; seen less on imaging	Thicker; seen more often on imaging

	Anterior lamina	Posterior lamina
Bounds	Perirenal space (behind) / anterior pararenal space (in front)	Perirenal space (in front) / posterior pararenal space (behind)

- **Lateroconal fascia:** formed where the two laminae **merge laterally**. It **separates the anterior and posterior pararenal spaces** and continues anterolaterally deep to the transversalis fascia.
- **Perirenal space contents:** **adrenal, kidney, ureter, perirenal fat, renal vascular pedicle, gonadal vessels**. Gerota's fascia extends inferiorly to envelop the ureter and gonadal vessels (Figs. 78.19 and 78.20, from Tobin, 1944).

PERIRENAL VERSUS PARARENAL FAT

Perirenal (perinephric) fat	Pararenal (paranephric) fat
Finer, lighter yellow; inside Gerota's fascia	Coarser, yellow-orange; outside Gerota's fascia

- **Use of the fat colour:** the difference in colour **guides the plane during colon mobilisation**.
- **Medial extent of the perirenal space:** the right and left spaces were once thought separate. In vivo cases and cadaveric injection studies suggest **communication across the midline below the renal hilum**.
- **Caudal extent:** once thought closed inferiorly by fusion of Gerota's fascia. It is in fact **cone-shaped and open inferiorly into the extraperitoneal pelvis**.
- **Clinical consequence:** these boundaries **contain perinephric collections** but let them **track caudally**:
 - **Urine:** traumatic or iatrogenic extravasation, or calyceal rupture in obstruction.
 - **Blood:** traumatic or iatrogenic perinephric haematoma, or ruptured aneurysm.
 - **Pus:** perinephric abscess or infected urinoma.

Anterior Pararenal Space and Inner Stratum

- **Boundaries:** the **anterior lamina of the renal fascia** behind and the **posterior parietal peritoneum** in front.
- **Surgical use:** the space is developed **medially from the white line of Toldt** to reach the kidney anteriorly.
- **White line of Toldt:** the **lateral border of the fusion of the colonic mesentery with the posterior peritoneum**. It forms when the inner stratum fuses with the primary dorsal peritoneum during gut rotation and fixation.
- **Contents:** the **secondarily retroperitoneal organs**, namely ascending and descending colon, pancreas, and the **second and third parts of the duodenum**. Each was intraperitoneal until the inner stratum fused posteriorly.

Figures

- **Fig. 78.18, fasciae and fat around the kidney:** the layers run from anterolateral wall muscles → transversalis fascia → paranephric fat → Gerota's fascia → perinephric fat → kidney, with psoas, quadratus lumborum and the IVC. From *Gray's Anatomy for Students*, 2010.

KEY POINTS: RETROPERITONEAL FASCIAE AND SPACES

- **Posterior pararenal space:** **posterior lamina of Gerota's fascia** in front; **transversalis fascia** behind and laterally.
- **Perirenal space:** bounded by the **anterior and posterior laminae** of Gerota's fascia.
- **Perinephric collections spread caudally** through the inferior opening of the perirenal space.

- **White line of Toldt:** the lateral border of fusion of the colonic mesentery with the posterior peritoneum.

Concept — Each space is the **territory of one embryological stratum**: outer for the body wall, intermediate for the genitourinary organs, inner for the gut. The **open inferior apex of the perirenal cone** explains why perinephric urine, blood and pus track into the pelvis.

GASTROINTESTINAL VISCERA AND MESENTERY

- **Non-urological retroperitoneal organs:** the **pancreas** and parts of the **duodenum and colon** (Figs. 78.23 and 78.24).

GASTROINTESTINAL VISCERA: RELATIONS AND OPERATIVE HAZARDS

Structure	Relation	Hazard / manoeuvre
Pancreatic head	Anterior to IVC ; cradled by the 2nd part of the duodenum	Injury in right renal surgery
Pancreatic neck and body	Body crosses anterior to the aorta and superior mesenteric artery (SMA) origin	—
Pancreatic tail	With the spleen , near the left upper pole and left adrenal	Injury in left retroperitoneal surgery
Stomach	Anterior to the left upper pole	Account for it in transperitoneal left renal surgery
Duodenum D1 (superior)	Intraperitoneal ; pylorus → neck of gallbladder	—
Duodenum D2 (descending)	Retroperitoneal; next to the right renal hilum; ampulla of Vater (common bile duct and main pancreatic duct)	Kocher manoeuvre mobilises it medially. It may be met on the left as the colon is reflected and the mesentery thins
Duodenum D3 (horizontal)	Crosses right → left, between SMA (in front) and aorta (behind)	—
Duodenum D4 (ascending)	Becomes intraperitoneal at the jejunum	—
Ascending colon, hepatic flexure	Overlie right retroperitoneal structures	Reflect medially at the white line of Toldt
Splenic flexure, descending colon	Overlie left retroperitoneal structures	Reflect medially at the white line of Toldt

- **Duodenal length and parts:** the duodenum is **20–25 cm** in four parts. **Only D2 and D3 are retroperitoneal.**
- **Transperitoneal access to the kidney:** the **ipsilateral colon is reflected medially** at the white line of Toldt, the **transition from colonic visceral to posterior parietal peritoneum.**
- **Hepatocolic and splenocolic ligaments:** **divide them sharply** to avoid liver and splenic injury. That injury is usually caused by **excessive retraction.**

The Mesentery as an Organ

- **Current view:** the mesentery is now classed as a **distinct organ** with intestinal, vascular, immunological and endocrine functions.

- **History: Leonardo da Vinci** drew it as continuous. A **fragmented model** was adopted in the late 1800s, with no mesentery for the ascending or descending colon. An observational cohort of total abdominal colectomies **re-established continuity from the ileocaecal to the rectosigmoid level**.
- **Functions:**
 - **Scaffold** suspending the bowel from the posterior wall.
 - **First barrier** between gut and body, and a **sampling reservoir for systemic immunity**. Mesenteric nodes regulate cell-mediated responses with the mucosa.
 - Carries **postganglionic nerves** from the main abdominal ganglia. Its role in the enteric nervous system is poorly understood.
- **Urological relevance:**
 - **Internal hernia** through a surgically created mesenteric defect is potentially lethal. **Close small mesenteric defects primarily** after bowel isolation, because small defects carry the higher herniation risk.
 - **Vascular mesenteropathy** can cause **loss of the bowel segment** used for diversion, ureteric interposition or augmentation.

Figures

- **Fig. 78.24, CT cross-sections, cephalad → caudal:**
 - (A) upper poles, above the pedicles
 - (B) **renal arteries and veins**
 - (C) renal pelves, with the **duodenum against the right hilum**
 - (D) lower poles and upper ureters.
- **Fig. 78.23:** colon, duodenum (four parts) and pancreas in situ with the kidneys, adrenals, portal vein and superior mesenteric vessels. From *Gray's Anatomy for Students*, 2010.

Concept — The bowel that must be moved to reach the kidney is **side-specific**. On the **right** it is the **duodenum (Kocher) and pancreatic head**; on the **left** it is the **pancreatic tail, spleen and stomach**. On both sides the colon is reflected at the white line of Toldt.

VASCULATURE

Arterial System

- **Arterial wall, three layers:**
 - **Intima:** endothelium plus subendothelial connective tissue.
 - **Media:** smooth muscle and elastic tissue that **control calibre**, bounded by the internal and external elastic laminae.
 - **Adventitia:** carries the vasomotor nerves and the **vasa vasorum** (“vessels of the vessels”).
- **Great vessel compartment:** runs from **T12 to L4/5**, between the perirenal spaces.
- **Aorta:** enters through the **aortic hiatus at T12** and runs centrally, **left of the IVC**.

TABLE 78.3 BRANCHES OF THE ABDOMINAL AORTA

Artery	Type	Origin / level	Supplies
Inferior phrenic	Paired lateral	First branches , just below the aortic hiatus	Diaphragm; superior adrenal branch
Celiac trunk	Anterior, unpaired	Just below the hiatus, T12	Foregut: left gastric, splenic, common hepatic

Artery	Type	Origin / level	Supplies
Middle adrenal	Paired lateral	Just above renal arteries (at SMA level)	Adrenals
SMA	Anterior, unpaired	Below celiac, L1–L2	Midgut: pancreas, small bowel, most of the large bowel
Renal	Paired lateral	Just below SMA, L1–L2	Kidneys; inferior adrenal branch
Gonadal (testicular / ovarian)	Paired anterolateral	Below renal arteries	Testes / ovaries
Lumbar	Paired posterior	Usually four pairs , beside L1–L4 bodies	Posterior body wall, spine and spinal cord
Inferior mesenteric (IMA)	Anterior, unpaired	Below renal arteries, L3–L4	Hindgut: splenic flexure → upper rectum
Median sacral	Posterior, unpaired	Just above the bifurcation; over L5, — sacrum and coccyx	—
Common iliac	Terminal	Bifurcation at L4	No named branches before dividing into internal and external iliac

Modified from Drake, Vogl and Mitchell, *Gray's Anatomy for Students*, 2005. Vertebral levels from the body text.

- **What the table is for:** it gives the **order of origin** and the **anterior/lateral/posterior pattern**.
- **Pattern:** **unpaired anterior branches supply the gut** (foregut → midgut → hindgut). **Paired lateral branches supply the retroperitoneal organs** (adrenals, kidneys, gonads) and share their lymphatic drainage. Posterior branches supply the wall.
- **Trap:** the gonadal arteries arise **anterolaterally below the renals**, not laterally. They occasionally arise from the renal artery, and then run with the gonadal vein.
- **Adrenal arterial supply has three sources:**
 - **Superior** from the inferior phrenic artery. This supply is **constant**.
 - **Middle and inferior** are variable. The commonest pattern is the middle adrenal from the **aorta** and the inferior adrenal from the **renal artery**.
- **Celiac trunk:** short, unpaired and anterior. It supplies the abdominal oesophagus, stomach, duodenum, spleen, liver and pancreas. The **splenic vessels run along the cephalad edge of the pancreatic body and tail**, so they are at risk when the **inferior pancreatic edge is lifted off the anterior renal fascia** in transperitoneal adrenal or renal surgery.
- **SMA:** branches are the inferior pancreaticoduodenal, ileocolic, right colic and middle colic arteries. **SMA injury in left-sided surgery → severe bowel ischaemia**, despite the collaterals.
- **Marginal artery of Drummond:** the **middle colic (SMA) ↔ left colic (IMA)** anastomosis. It is the **SMA-to-IMA collateral** that lets the IMA be sacrificed without colonic ischaemia.
- **Renal arteries:** arise at **L1–L2**. **At least a quarter of people have supernumerary renal arteries**. There is **no consensus on side predominance**, and female renal arteries are **smaller** on CT angiography. Variants are covered in Chapter 84.
- **Testicular arteries:** run anterior to psoas, the genitofemoral nerve and the ipsilateral ureter (and, on the right, the IVC) towards the **internal inguinal ring**.
- **Ovarian arteries:** run anterior to the ureter and then medially through the **infundibulopelvic (suspensory) ligament**.
- **Lumbar arteries:** a **fifth pair** sometimes arises from the median sacral artery.

- **IMA:** branches are the left colic, sigmoid (→ two to three inferior left colic arteries) and superior haemorrhoidal (rectal) arteries. The **superior rectal artery** collateralises with the **middle and inferior rectal arteries** from the internal iliac territory, so the rectum survives IMA ligation.
- **Ureteric blood supply:**
 - **Proximal:** renal artery.
 - **Middle:** aorta, with contributions from the common iliac, gonadal, uterine, middle rectal and vaginal arteries.
 - **Distal:** internal iliac artery, including the superior and inferior vesical arteries.
- **Direction of ureteric supply:** the **abdominal ureter is supplied medially** and the **pelvic ureter laterally**.

VESSELS THAT CAN BE SACRIFICED, AND WHY

Vessel	Collateral that protects the end organ
IMA (colon)	Marginal artery of Drummond (middle colic ↔ left colic)
IMA (rectum)	Superior rectal ↔ middle and inferior rectal (internal iliac territory)
Testicular / ovarian arteries	Extensive gonadal collaterals in both sexes
Median sacral artery	No end-organ ischaemia
Contrast: SMA	Not safely sacrificed; injury → severe bowel ischaemia

Venous System

ARTERIAL VERSUS VENOUS WALL

	Artery	Vein
Elastic laminae	Well defined	Poorly defined , even in large veins
Media	Thick , more smooth muscle	Thin , less smooth muscle
Adventitia	Carries nerves and vasa vasorum	Larger than the media
Valves	None	Bicuspid valves prevent retrograde flow

- **IVC course:** formed by the **confluence of the common iliac veins below and to the right of the aortic bifurcation**. It ascends anterior to the vertebral bodies, **right of the aorta**, running alongside it below the renal veins and becoming more anterior as it rises.
- **At the diaphragm** the **right crus** separates the IVC from the aorta. The IVC passes through the **central tendon at T8** into the right atrium.
- **Venous variability:** veins vary more than arteries, although most parallel their arteries.
- **Median sacral vein:** usually drains to the **left common iliac vein**, sometimes into the angle between the common iliacs. **Avoid it when fixing mesh in sacrocolpopexy.**
- **Ascending lumbar veins:** run **behind psoas, lateral to the spine**. They link with variable lumbar veins, sometimes as a plexus in front of the vertebral bodies, and continue as the **hemiazygos (left) and azygos (right)** veins.

LEFT–RIGHT VENOUS ASYMMETRY

Vein	Left	Right
Gonadal (testicular / ovarian)	Into the left renal vein at a right angle ; occasionally into the IVC; longer	Into the anterolateral IVC ; into the right renal vein in up to 10%
Renal vein	Long ; receives the gonadal (caudal), adrenal (superior) and ≥1 lumbar vein (near the gonadal ostium)	Short , usually no tributaries (rarely gonadal or lumbar); duplicated in one-sixth

Vein	Left	Right
Adrenal	Superior margin of the left renal vein, just medial to the gonadal ; may be joined by the left inferior phrenic	Short, single , no tributaries, straight into the posterolateral IVC
Inferior phrenic	May join the left adrenal vein	Usually into the upper IVC

- **Renal veins:** lie **anterior to the renal arteries** and enter the lateral IVC at **L1**.
- **Left varicocele predominance:** the **longer left testicular vein and its perpendicular entry** into the left renal vein raise back pressure. Pampiniform plexus → testicular veins, which ascend lateral to their artery and anterior to the ureter.
- **Sudden right varicocele → image the retroperitoneum** to exclude obstructing malignancy, for example **right renal cell carcinoma with venous thrombus**.
- **Right-sided ureteric identification:** develop the plane **lateral to the gonadal vein**, leaving the vein medial. A plane **medial** to it risks **IVC injury**, and lifting the kidney then risks **gonadal vein avulsion**.
- **Ovarian veins:** drain the pampiniform plexus at the ovarian hilum via the infundibulopelvic ligament. The **same left/right pattern** as in males applies.

Portal and Hepatic Venous Drainage

- **Portal system:** collects from bowel, spleen, pancreas and gallbladder and drains into the liver (Fig. 78.35). It **does not mirror the arteries** as closely as the systemic veins do.
- **Superior mesenteric vein (SMV):** drains the **small bowel and colon proximal to the splenic flexure**. Tributaries are the right gastroepiploic, anterior and posterior inferior pancreaticoduodenal, jejunal, ileal, ileocolic, right colic and middle colic veins.
- **Portal vein formation:** **SMV + splenic vein → portal vein**. Splenic tributaries are the inferior mesenteric, short gastric, left gastroepiploic and pancreatic veins.
- **Inferior mesenteric vein (IMV):** drains the **colon distal to the splenic flexure**. It joins the **SMV rather than the splenic vein in 40%**. It lies in the mesocolon when the left colon is reflected.
- **Reoperative trap:** the **IMV can be mistaken for the gonadal vein** when the mesentery is stuck to Gerota's fascia.
- **Hepatic outflow:** portal vein → right and left branches → **hepatic sinusoids** → hepatic veins (a **larger upper** and a **smaller lower group**) → **anterior IVC** below the diaphragm.
- **Budd–Chiari syndrome:** **hepatic vein occlusion** causing progressive, often rapid, liver failure with **jaundice, ascites, abdominal pain and hepatomegaly**.

Figures

- **Figs. 78.28 and 78.29:** CT angiogram of the aorta, with hepatic, splenic, renal, inferior mesenteric and iliac arteries, and a diagram of the **IVC and its tributaries alongside the aorta and its branches**. Fig. 78.28 from *Hinman's Atlas of Urosurgical Anatomy*, 2012.
- **eFigs. 78.26 and 78.27:** arterial wall structure and the aorta with its branches. From *Hinman's Atlas of Urosurgical Anatomy*, 2012.

KEY POINTS: RETROPERITONEAL VASCULATURE

- **Adrenal gland arterial supply:** inferior phrenic artery, aorta and renal artery.
- **Renal hila at L1; renal veins anterior to renal arteries.**
- **Gonadal arteries** arise anterolaterally from the aorta **below the renal arteries**.
- **Marginal artery of Drummond** (SMA → IMA) allows **IMA sacrifice without colonic ischaemia**.

- **Proximal ureter supplied medially, distal ureter laterally.**
- **Left gonadal vein → left renal vein at a right angle**, hence more left varicoceles. **Right gonadal vein → IVC.**

Concept — The arterial tree follows **embryological territory**: unpaired anterior branches to the gut, paired lateral branches to the retroperitoneal organs. Venous anatomy is dominated by **left-right asymmetry** because the **IVC lies on the right**. Every long left vein must cross to reach it, and that single fact explains the left renal vein's tributaries, left varicoceles, and the warning carried by a new right varicocele.

LYMPHATIC SYSTEM

- **Node structure**: nodes have **multiple afferent** and a **single efferent** lymphatic.
- **Direction of flow**: lymph flows **cephalad and right → left**, returning to the **left innominate (brachiocephalic) vein**.
- **Exception to the left-sided return**: the head, neck, right thorax, right arm and right heart drain to the **right innominate vein**.
- **Pelvic and lower limb drainage**: internal iliac, external iliac, common iliac, obturator and sacral nodes → **lumbar nodes → lumbar trunks**.
- **Why urologists care about lumbar nodes**: they are the **primary drainage of organs supplied by the lateral aortic branches**, namely the **kidneys, adrenals, ureters and gonads** (Fig. 78.36).

LUMBAR NODAL GROUPS

Group	Subgroups	Territory
Left lumbar (aortic)	Preaortic (celiac, superior mesenteric, inferior mesenteric → intestinal trunk); left para-aortic ; retroaortic (variable)	Para-aortic region runs from the aortic midline to the left ureter . Retroaortic nodes lie between the aorta and the vertebrae
Interaortocaval (interaortocaval)	—	IVC midline → aortic midline
Right lumbar (caval)	Precaval, right paracaval, retrocaval	Paracaval region runs from the IVC midline to the right ureter . Retrocaval nodes lie between the IVC and psoas

Testicular Lymphatic Drainage

- **Why the testis drains here**: it is **embryologically retroperitoneal**, with a retroperitoneal blood supply and primary lymphatic drainage.
- **Three key regions** for testicular malignancy: **left para-aortic, interaortocaval, right paracaval**.

LEFT VERSUS RIGHT TESTIS: PRIMARY LANDING ZONES

	Left testis	Right testis
Primary	Left para-aortic	Interaortocaval
Secondary	Interaortocaval	Right paracaval
Crossover	None significant to right paracaval	Small but appreciable to left para-aortic

- **What the table tests**: crossover is **one-way, right → left**, which matches the global right-to-left flow of lymph.
- **Cisterna chyli and thoracic duct**: the lumbar trunks meet at the **cisterna chyli**, which is **behind the right side of the aorta and in front of L1-L2**. This is the start of the **thoracic duct**, which ascends **behind the aorta** to the **left innominate vein**.

KEY POINTS: RETROPERITONEAL LYMPHATIC SYSTEM

- **Testes are embryologically retroperitoneal**, with retroperitoneal blood supply and primary lymphatic drainage.
- **Left testis → left para-aortic**, with some drainage to the interaortocaval nodes.
- **Right testis → interaortocaval**, with some drainage to the right paracaval nodes and a small amount to the left para-aortic nodes.
- Testicular drainage follows the **global right-to-left lymphatic flow**.

Concept — The testis drains to the renal hilum because it **began there**. Lymph drifts **right to left**, so a right tumour can reach left-sided nodes but a left tumour does not reach right-sided ones.

NERVOUS STRUCTURES**AUTONOMIC VERSUS SOMATIC**

Autonomic	Somatic
Efferent and afferent supply to viscera, vessels and smooth muscle	Efferent and afferent supply to skeletal muscle, skin and peritoneum

Autonomic Nervous System

- **Two-neuron chain:**
 - Preganglionic neuron**, with its cell body in the central nervous system
 - synapses in a **ganglion**
 - **postganglionic neuron**, whose axon enters the target organ.
- **Adrenal exception: preganglionic fibres synapse directly on the adrenal medulla** and trigger catecholamine release. The adrenal is effectively a **specialised sympathetic ganglion**.

PARASYMPATHETIC VERSUS SYMPATHETIC

	Parasympathetic	Sympathetic
Outflow	Craniosacral: cranial nerves III, VII, IX, X and S2–S4	Thoracolumbar: spinal cord segments T1–L2
Pathway	S2–S4 → pelvic splanchnic nerves to pelvic and abdominal viscera; vagus to thoracic, abdominal and pelvic viscera	Ventral root → spinal nerve → anterior ramus → ipsilateral sympathetic trunk
Ganglion site	Postganglionic fibres usually within the wall of the target organ	Sympathetic trunk ganglia (→ body wall, lower limbs), or via splanchnic nerves to the aortic autonomic plexuses
“Splanchnic” nerves	Pelvic splanchnics only	Thoracic (greater, lesser, least), lumbar and sacral splanchnics

- **Sympathetic trunks:** run **medial to psoas** on the anterolateral spine. The **lumbar arteries and veins cross them at right angles**.
- **Celiac plexus** is the **first and largest**, with paired ganglia lateral to the celiac artery. It carries much of the autonomic supply to the **kidney, adrenal, renal pelvis and ureter**, and **some testicular fibres**, which then run with the testicular artery.
- **Renal plexus:** continuous with the celiac plexus and lies beside the renal arteries. It contains the **aorticorenal ganglion**, an inferior extension of the celiac ganglion.

- **Superior and inferior hypogastric plexuses** (contiguous) carry much of the **sympathetic supply to the pelvic viscera**. The superior plexus runs from the **caudal aorta to the front of L5**.
- **Hypogastric plexus injury**: extensive retroperitoneal dissection that disrupts these plexuses causes **loss of seminal emission** or **failure of bladder neck closure** → **retrograde ejaculation**.

Somatic Nervous System

- **Lumbosacral plexus**: formed from the **anterior rami of the lumbar and sacral nerves plus T12**, in the retroperitoneum. It supplies the lower abdomen and lower limbs.
- **Relation to psoas**: **superior branches pierce psoas; inferior branches run medial to it** (Fig. 78.40).
- **Subcostal (T12), iliohypogastric (L1) and ilioinguinal (L1)**: the subcostal runs below the 12th rib. All three cross the **front of quadratus lumborum**, pierce transversus and run **deep to internal oblique**. They supply the abdominal wall muscles and the skin of the posterolateral buttock, upper medial thigh and genitalia.
- **Genitofemoral (L1–L2)**: runs **on the anterior surface of psoas, parallel to it**, and divides near the inguinal ligament.
 - **Femoral branch**: passes under the inguinal ligament into the femoral sheath and supplies the **upper anterior thigh**.
 - **Genital branch**: enters the deep ring. It is **motor to cremaster (the efferent limb of the cremasteric reflex)** and sensory to the anterior scrotum, or to the mons and labium majus.
- **Lateral cutaneous nerve of the thigh (L2–L3)**: sensory to the anterior and lateral thigh.
- **Obturator (L2–L4)**: arises behind psoas and runs to the obturator canal. It is **motor to the thigh adductors**.
- **Femoral (L2–L4)**: motor to the **anterior thigh (knee extension)** and to **iliacus and pectineus (hip flexion)**. Sensory to the **anteromedial lower limb**.
- **Sciatic (L4–S3)**: most of the motor and sensory supply to the lower limb, including the **posterior thigh and all leg and foot muscles**.

TABLE 78.4 BRANCHES OF THE LUMBOSACRAL PLEXUS

Nerve	Segments	Motor	Sensory
Subcostal	T12	Abdominal wall muscles	Skin over the hip
Iliohypogastric	L1	Internal oblique, transversus	Posterolateral gluteal and pubic skin
Ilioinguinal	L1	Internal oblique, transversus	Upper medial thigh; root of penis and anterior scrotum / mons and labium majus
Genitofemoral	L1–L2	Genital branch: cremaster	Genital branch: anterior scrotum / mons and labium majus. Femoral branch: upper anterior thigh
Lateral cutaneous nerve of thigh	L2–L3	None (purely sensory)	Anterior and lateral thigh to the knee
Obturator	L2–L4	Obturator externus, pectineus, medial (adductor) compartment	Medial thigh
Femoral	L2–L4	Iliacus, pectineus, anterior compartment	Anterior thigh, medial leg (saphenous)

Modified from Drake, Vogl and Mitchell, *Gray's Anatomy for Students*, 2005.

- **What the table is for**: it gives the segmental origin plus motor and sensory territory of each nerve at risk in urological surgery.
- **Pattern**: the root level falls one step at a time, **T12 → L1 → L1–L2 → L2–L3 → L2–L4**, and territory moves from trunk → groin → thigh.

- **Trap:** both the ilioinguinal *and* the genital branch of genitofemoral supply the **anterior scrotum**, but only the **genitofemoral is motor to cremaster**. The **lateral cutaneous nerve of the thigh has no motor function**.
- **Pectineus overlap:** it is supplied by **both femoral and obturator** nerves.

NERVE INJURY IN UROLOGICAL SURGERY AND POSITIONING

Nerve	Setting	Mechanism	Result
Genitofemoral	Psoas hitch; laparoscopic varicocelectomy	Suture placement; ligation	Anterior scrotal / upper anterior thigh sensory loss
Obturator	Transurethral resection of bladder tumour (TURBT) on the lateral wall; pelvic lymph node dissection	Electrocautery stimulates the nerve	Violent hip adduction → bladder perforation
Femoral	Retractor blades against the inguinal ligament; prolonged hip flexion in low lithotomy	Compression; stretch	Quadriceps palsy, loss of knee extension
Sciatic	High lithotomy (vaginal and urethral surgery)	Prolonged hip hyperflexion	Posterior thigh, leg and foot deficit
Hypogastric plexuses	Extensive retroperitoneal dissection	Plexus disruption	Loss of emission; retrograde ejaculation

Figures

- **Fig. 78.37, sympathetic chain and splanchnic nerves:** cervical → thoracic → lumbar → sacral ganglia, ending at the **ganglion impar**. It shows greater, lesser and least thoracic splanchnics, lumbar and sacral splanchnics, and pelvic splanchnics joining the inferior hypogastric plexus. From *Gray's Anatomy for Students*, 2010.
- **Fig. 78.38, prevertebral plexuses:** celiac (with celiac and aorticorenal ganglia) → aortic → superior hypogastric → hypogastric nerves → inferior hypogastric, with superior and inferior mesenteric ganglia. From *Gray's Anatomy for Students*, 2010.
- **Fig. 78.41, cutaneous distribution of the lumbar plexus:** T10–L1 bands on the trunk, **ilioinguinal (L1) to the groin and genitalia**, lateral cutaneous nerve to the lateral thigh, obturator to the medial thigh, and femoral (intermediate and medial cutaneous, **saphenous**) to the anterior thigh and medial leg. From *Gray's Anatomy for Students*, 2010.

KEY POINTS: NERVOUS STRUCTURES

- **Parasympathetic outflow is craniosacral;** postganglionic fibres are often in the wall of the viscus.
- **Sympathetic preganglionic fibres leave T1–L2** and synapse in the **sympathetic trunk or the autonomic plexuses**.
- The **lumbosacral plexus** gives somatic sensory and motor supply to the pelvis and lower limbs.

Concept — The nerves at risk are those **lying on or in psoas and the pelvic side wall**. Each nerve's injury follows directly from its **motor territory**: adduction for obturator, knee extension for femoral, cremaster for genitofemoral, and emission and bladder neck closure for the hypogastric plexus.